

CAAM/STAT31470: Homework #1

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: In this week's homework we will look at some simple examples of functions of 1 complex variable. In particular, we will look at how they map values, and how to compute limits and derivatives of these functions.

Homework is due: **Wednesday Jan 15, 5pm.**

Problem 1.1 (8 points)

- (a) Sketch the region onto which the sector

$$S = \{r \leq 2, 0 \leq \theta \leq \pi/4\}$$

is mapped by the transformations: (i) $w = z^2$; (ii) $w = z^3$; and (iii) $w = z^4$.

- (b) Show that the lines $ay = x$ (with $a \neq 0$) are mapped onto the spirals $\rho = \exp(a\phi)$ under the transformation $w = \exp z$ where $w = \rho \exp(i\phi)$.

Problem 1.2 (6 points)

One interpretation of a function $w = f(z) = u(x, y) + iv(x, y)$ is that of a vector field in the domain of definition of f . The function assigns a vector w , with components $u(x, y)$ and $v(x, y)$, to each point z at which it is defined. Indicate graphically the vector fields represented by (a) $w = iz$; and (b) $w = z/|z|$.

Problem 1.3 (8 points)

Using the definition of limits, show that

- (a) $\lim_{z \rightarrow z_0} \operatorname{Re} z = \operatorname{Re} z_0$
- (b) $\lim_{z \rightarrow 0} \bar{z}^2/z = 0$
- (c) $\lim_{z \rightarrow z_0} (z/\bar{z})^n$ does not exist.
- (d) $\lim_{z \rightarrow \infty} 4z^2/(z-1)^2 = 4$

Problem 1.4 (8 points)

- (a) Let f be defined by

$$(1) \quad f(z) = \begin{cases} \bar{z}^2/z, & z \neq 0 \\ 0 & z = 0 \end{cases}$$

Is f differentiable at $z = 0$?

- (b) Suppose that $f(z_0) = g(z_0) = 0$, and $f'(z_0)$ and $g'(z_0)$ exist. Show that

$$(2) \quad \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$$

End of homework